

Appendix D

The Matrix Computation Toolbox

The Matrix Computation Toolbox is a collection of MATLAB M-files containing functions for constructing test matrices, computing matrix factorizations, visualizing matrices, and carrying out direct search optimization. Various other miscellaneous functions are also included.

This toolbox supersedes the Test Matrix Toolbox [606, 1995] that was described in the first edition of this book. Most of the test matrices in that toolbox have been incorporated into MATLAB in the `gallery` function. The new toolbox incorporates some of the other routines in the Test Matrix Toolbox and adds several new ones.

Of particular relevance to this book are functions for

- Cholesky factorization with complete pivoting for a symmetric positive semidefinite matrix: `cholp`;
- rounding matrix elements to a specified number of bits—useful for simulating lower precision in experiments: `chop`;
- GE with one of four optional pivoting strategies: no pivoting, partial pivoting, rook pivoting, and complete pivoting: `gep`;
- Gauss–Jordan elimination with no pivoting or partial pivoting for solving a linear system: `gj`;
- generalized QR factorization: `gqr`;
- the classical and modified Gram–Schmidt methods for QR factorization: `gs_c`, `gs_m`;
- block LDL^T factorization of symmetric matrices with partial pivoting or rook pivoting: `ldlt_symm`;
- block LDL^T factorization of symmetric tridiagonal matrices with Bunch’s pivoting strategy: `ldlt_sytr`;
- block LDL^T factorization of skew-symmetric matrices with Bunch’s partial pivoting strategy: `ldlt_skew`;
- solving the equality constrained least squares problem: `lse`;
- two- and three-dimensional views of pseudospectra: `ps`, `pscont`;

- direct search optimization by the alternating directions method, **ad**, the multidirectional search method, **mds**, and the Nelder–Mead simplex method, **nms**;
- the p -norm of a matrix: **pnorm**;
- Strassen’s method for matrix multiplication and Winograd’s variant: **strassen**, **strassenw**;
- converting between a triangular matrix and the nontrivial elements of its vec (useful in direct search optimization): **treshape**;
- the vec-permutation matrix: **vecperm**.

The Matrix Computation Toolbox is available from

<http://www.ma.man.ac.uk/~higham/mctoolbox>

We summarize the contents of the toolbox in the following tables, which list the M-files by category, with short descriptions.

Demonstration	
mctdemo	Demonstration of Matrix Computation Toolbox.

Test Matrices	
augment	Augmented system matrix.
gfpp	Matrix giving maximal growth factor for Gaussian elimination with partial pivoting.
makejcf	A matrix with specified Jordan canonical form.
rschur	An upper quasi-triangular matrix.
vand	Vandermonde matrix.
vecperm	Vec-permutation matrix.

Factorizations and Decompositions	
cholp	Cholesky factorization with complete pivoting of a positive semidefinite matrix.
cod	Complete orthogonal decomposition.
gep	Gaussian elimination with pivoting: none, complete, partial, or rook.
gj	Gauss–Jordan elimination with no pivoting or partial pivoting to solve $Ax = b$.
gqr	Generalized QR factorization.
gs_c	Classical Gram–Schmidt QR factorization.
gs_m	Modified Gram–Schmidt QR factorization.
ldlt_skew	Block LDL^T factorization for a skew-symmetric matrix.
ldlt_symm	Block LDL^T factorization with partial pivoting or rook pivoting for a symmetric indefinite matrix.
ldlt_sytr	Block LDL^T factorization for a symmetric tridiagonal matrix.
matsignt	Matrix sign function of a triangular matrix.
poldec	Polar decomposition.
signm	Matrix sign decomposition.
trap2tri	Unitary reduction of trapezoidal matrix to triangular form.

Visualization	
fv	Field of values (or numerical range).
gersh	Gershgorin disks.
ps	Dot plot of a pseudospectrum.
pscont	Contours and colour pictures of pseudospectra.
see	Pictures of a matrix.

Direct Search Optimization	
adsmx	Alternating directions method.
mdsmx	Multidirectional search method.
nmsmx	Nelder–Mead simplex method.

Miscellaneous	
chop	Round matrix elements.
cpltaxes	Determine suitable axis for plot of complex vector.
dual	Dual vector with respect to Hölder p -norm.
lse	Solve the equality constrained least squares problem.
matrix	Matrix Computation Toolbox information and matrix access by number.
pnorm	Estimate of matrix p -norm ($1 \leq p \leq \infty$).
rootm	P th root of a matrix.
seqcheb	Sequence of points related to Chebyshev polynomials.
seqm	Multiplicative sequence.
show	Display signs of matrix elements.
skewpart	Skew-symmetric (skew-Hermitian) part.
sparsify	Randomly set matrix elements to zero.
strassen	Strassen's fast matrix multiplication algorithm.
strassenw	Strassen's fast matrix multiplication algorithm (Winograd variant).
sub	Principal submatrix.
symmpart	Symmetric (Hermitian) part.
trshape	Reshape vector to or from (unit) triangular matrix.